

Hulhumalé: Assessing Land Reclamation as a Climate Adaptation Strategy in the Maldives

Despite its picture-perfect beaches and sparkling turquoise water, not everything in the Maldives is paradise. As the world's lowest-lying country with an average of 1.6 meters above sea level, the Maldives is particularly vulnerable to climate change-related coastal flooding and sea level rise.¹ Climate scientists predict that by the year 2100, most of the country's inhabited islands will be underwater, including Malé, the country's capital with over 100,000 inhabitants.² This environmental issue poses an existential threat that needs to be addressed urgently if the Maldives is to survive into the next century. One promising strategy that is already being implemented is land reclamation, the creation of artificial islands that are higher in elevation to withstand rising sea levels so that Maldivians can continue residing in their homeland. However, this strategy also brings a new set of social and environmental challenges for the country. Moreover, as climate change predictions continue to get increasingly bleak, land reclamation might only help stall the next disaster rather than serve as the silver bullet for the country's longevity.

The Maldives' most ambitious land reclamation project is Hulhumalé, a 400-hectare "smart city" on an artificial island 8 kilometers from Malé. This project is being led by the Housing Development Corporation (HDC), an entity owned by the Maldivian government. The land reclamation for this project was completed in 2015 and cost the HDC 160 million USD, most of which was borrowed from foreign banks.³ What makes Hulhumalé an attractive solution is that it is situated two meters above sea level with the expectation that the sea level will only

¹ Astaiza, Randy. "11 Islands That Will Vanish When Sea Levels Rise." Business Insider. October 12, 2012.

² McConnell, Tristan. "The Maldives Is Being Swallowed by the Sea. Can It Adapt?" Environment. National Geographic, January 20, 2022.

³ Miller, Norman. "A New Island of Hope Rising from the Indian Ocean." BBC Travel. BBC, September 10, 2020.

rise half a meter by 2100.⁴ Hulhumalé seeks to “welcome people from all the Islands of the Maldives” and is expected to receive a quarter million Maldivians by 2035, many of whom will be climate refugees.⁵ To achieve this goal, the HDC is providing a wide range of housing options to accommodate almost any Maldivian; 60% of mid-range housing units will be sold under the set price ceiling, and social housing will also be available.⁶ This commitment to equity demonstrates that Hulhumalé is serious about protecting the most vulnerable members of society and providing everyday Maldivians with an optimistic climate future. As part of this climate future, Hulhumalé is positioned to become the Maldives’ first “smart” and “green” city through modern urban planning, civil engineering, and sustainable architectural techniques, such as intelligent traffic management systems,⁷ electric buses,⁸ and natural temperature regulation through intentional building and street orientations.⁹ The island plans to source a third of its energy from solar panels and provide ample green space for residents.¹⁰ The HDC’s emphasis on making Hulhumalé a sustainable city is essential to strengthening its climate resilience.

Despite the project’s promising benefits, negative social and ecological externalities exist. Climate refugeeism threatens the social and economic fabric of the Maldives as the current maritime lifestyle and culture cannot be replicated in an urban setting such as Hulhumalé. This shift affects people individually as they adapt their livelihoods to fit into their new environment. One Maldivian mother who moved her family to an apartment in Hulhumalé after leaving her native island expressed, “We are living in these towers because we don’t have a choice,”¹¹

⁴ HDC, “Hulhumalé”

⁵ Housing Development Corporation (HDC). “Hulhumalé - Most Ambitious Urban Development Project in Maldives.” Housing Development Corporation, 2018.

⁶ Miller, “A New Island of Hope”

⁷ HDC, “Planning and Design”

⁸ Miller, “A New Island of Hope”

⁹ Díaz, Gabriela. “Hulhumalé, Maldives: Monumental Climate Adaptation Effort Raises Hope in a Threatened Paradise.” Global Center on Adaptation, February 25, 2022.

¹⁰ Miller, “A New Island of Hope”

¹¹ McConnell, “The Maldives Is Being Swallowed by the Sea.”

underscoring the lack of agency people feel regarding their relocation. On the ecological side, inherent environmental decay still results from land reclamation in such a biodiverse ecosystem. The displacement of millions of cubic meters of sand from the seafloor leads to coral reef degradation, thus decreasing the surrounding biodiversity and making islands more vulnerable to tropical storms since reefs act as a protective barrier. This was proven during the 2004 Indian Ocean Tsunami since islands that were surrounded by more coral reefs experienced less damage than islands that had less coral reef coverage.¹² With climate change expected to increase storm intensity and frequency, coral reefs are an essential resource for disaster mitigation. Thus, land reclamation as a climate adaptation strategy is a double-edged sword. On the one hand, it brings the social benefit of allowing communities to flourish in their homeland; on the other hand, it leaves these same communities vulnerable to other environmental disasters.

Although Hulhumalé is an ambitious climate adaptation project, its biggest pitfall perhaps is that it might not be ambitious enough to guarantee its longevity. When this project was designed in the mid-1990s, scientists estimated that the global sea level would only rise half a meter by 2100, thus making Hulhumalé's two meters of elevation sufficient.¹³ Since then, the rate of global sea level rise has doubled¹⁴ and NASA estimates that sea levels in 2300 might be five meters higher than they are today if global emissions are not reduced.¹⁵ This is alarming because the United Nations stated in 2022 that the global community is failing to limit global warming to the 1.5°C stipulated in the Paris Agreement.¹⁶ The climate estimates used to plan Hulhumalé do not reflect the trajectory of climate change now, thus jeopardizing the project's long-term

¹² Kan et al., "2004 Indian Ocean Tsunami: Coral Reefs and Islands."

¹³ Miller, "A New Island of Hope"

¹⁴ Hamlington et al., "The rate of global sea level rise doubled during the past three decades." *Communications Earth & Environment*, October 17, 2024.

¹⁵ Voiland, Adam. "Anticipating Future Sea Levels." NASA. NASA, 2021.

¹⁶ UNFCCC. "Climate Plans Remain Insufficient: More Ambitious Action Needed Now." UNFCCC, October 26, 2022.

resiliency. As climate change accelerates due to governments' inability to curb carbon emissions, many climate adaptation measures could fail due to their reliance on conservative predictions.

One conspicuous solution for land reclamation in the Maldives is to create higher islands. The cost of land reclamation at Hulhumalé's current elevation is \$150,000 per hectare, which is noticeably lower than similar land reclamation projects in other countries.¹⁷ Given this relatively affordable figure, it is not unreasonable to suggest that the long-term benefits of building future islands three or four meters above sea level would be worth the additional cost, especially since developed countries have already agreed to increase funding for climate-adaptation projects in less developed countries.¹⁸ Although this option seems enticing, building higher islands would still degrade the surrounding ecosystem and worsen other forms of environmental vulnerability. Thus, while land reclamation can serve as a short-term solution to the climate refugee crisis, it is not a sustainable or resilient practice for island nations in the long run.

The most effective way to safeguard the resiliency of climate adaptation projects such as Hulhumalé is to target the root cause of the issue: climate change itself. If the international community were to properly enforce the Paris Agreement and limit global warming to 1.5°C, sea level rise would be a lesser threat to island nations, allowing strategies such as land reclamation to be more viable. Needless to say, this approach falls outside the scope of Hulhumalé and even the Maldives as it would require an enormous amount of international cooperation and investment. However, a global commitment to reducing anthropogenic climate change would allow vulnerable populations to maintain their livelihoods for centuries. The contradiction of this issue is that the countries that contribute the most to climate change are not usually the ones who

¹⁷ Sim, Chi Yin. "A Mirage of Luxury Built on Sand." Pulitzer Center, March 4, 2019. <https://pulitzercenter.org/stories/mirage-luxury-built-sand#:~:text=Reclaimed%20land%20usually%20costs%20between,buy%20land%2C%20than%20to%20reclaim.>

¹⁸ UNFCCC. "COP29 UN Climate Conference Agrees to Triple Finance to Developing Countries, Protecting Lives and Livelihoods" UNFCCC, November 24, 2024.

face the brunt of its consequences. For reference, the average Maldivian only emits 2.2 tons of CO₂ a year, one-eighth the emissions of the average American.¹⁹ Because of this imbalance, the Global North should bear a greater share of the responsibility to reduce greenhouse gas emissions as a form of global climate justice.

Although land reclamation could allow hundreds of thousands of climate refugees to remain in the Maldives, the subsequent social and economic shifts would be incredibly disruptive to their current livelihoods. While Hulhumalé seems to be an adequate solution for housing climate refugees in the short term, the Maldives needs a more promising long-term solution to continue flourishing. This case study highlights two fundamental ideas: the need for both adaptation and mitigation measures to address climate change and how Global North-South dynamics unequally distribute vulnerability. The reason why Hulhumalé will not last as long as intended is that climate change has since accelerated rather than slowed. Thus, climate adaptation is almost futile without mitigation. The Maldives cannot adapt its way out of climate change since every intervention carries its own set of negative environmental externalities. The Maldives also cannot mitigate its way out of the climate crisis since its emissions are insignificant on a global scale. Hence, Global South communities are burdened by a wicked problem they did not cause. To address this issue equitably, the Global North needs to provide more funding for adaptation projects in the Global South while also mitigating its greenhouse gas emissions. Through international collaboration on adaptation and mitigation measures, there is still hope for the Maldives to survive into the next century and live out a better climate future.

¹⁹ Knoema. "Maldives CO2 emissions, 1970-2021." Knoema, 2021.
<https://knoema.com/atlas/Maldives/CO2-emissions#:~:text=In%202021%2C%20CO2%20emissions%20for,average%20annual%20rate%20of%2015.70%25>.